THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. What would be the electron dot structure of carbon dioxide (CO_2)? The number of shared electron pairs in it is: 1

(a) 6 **(b)** 3 **(c)** 2 **(d)** 4

2. Soaps are sodium or potassium salts of: 1

(a) long-chain carboxylic acids **(b)** aldehydes
(c) alkanes **(d)** short-chain alcohols

3. The property of carbon to form long chains by bonding with itself is called: 1

(a) catenation **(b)** isomerism **(c)** allotropy **(d)** polymerisation

4. The number of covalent bonds in a molecule of pentane C_5H_{12} is: 1

(a) 14 **(b)** 17 **(c)** 15 **(d)** 16

5. The cleansing action of soap is due to: 1

(a) reaction with dirt **(b)** its smell
(c) its acidity **(d)** formation of micelles

6. Carbon forms covalent bonds because it has: 1

(a) four valence electrons **(b)** eight valence electrons
(c) two valence electrons **(d)** six valence electrons

7. Which of the following is an unsaturated hydrocarbon? 1

(a) C_2H_6 **(b)** C_3H_8 **(c)** CH_4 **(d)** C_2H_4

8. Soap does not lather well in hard water because of: 1

(a) low temperature **(b)** acidity
(c) calcium and magnesium salts forming scum **(d)** sodium chloride

9. The number of isomers of butane is: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

10. The next homologue of C_3H_6 is: 1

(a) C_4H_{10} **(b)** C_3H_8 **(c)** C_5H_{10} **(d)** C_4H_8

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does carbon form compounds mainly by covalent bonding? 2

12. What is a functional group? Name the one in ethanol. 2

13. What is esterification? Write the equation for ethanol and ethanoic acid. 2

14. What is catenation? Why is it more pronounced in carbon? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw the structures of the two isomers of butane and name them. 3

16. Explain the oxidation of ethanol to ethanoic acid. Name two oxidising agents. 3

17. Differentiate between saturated and unsaturated hydrocarbons with an example each. 3

OR

Write the molecular formula and structure of ethene. Why is it unsaturated? 3

18. What happens when ethanoic acid reacts with sodium hydroxide and with sodium carbonate? Write equations. 3

OR

Explain why diamond is hard and graphite is soft. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the bonding in carbon: four valence electrons, covalent bond, catenation and tetravalency. Draw dot structure of methane. 5

OR

What is esterification? Write the reaction, the conditions, the smell and uses of esters. Write the saponification reaction. 5

20. Explain structural isomerism with isomers of pentane: draw formulae and name them. 5

OR

Explain how soaps clean. Draw/describe a micelle and give the effect of hard water. 5

END OF QUESTION PAPER — Total: 40 Marks

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